MINI PROJECT REPORT

ON

# CampUs

Submitted by


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to

the APJ Abdul Kalam Technological University

in partial fulfillment of the requirements for the award of the degree

of

Bachelor of Technology

in

***Computer Science and Engineering***


**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

**ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI**

**(AUTONOMOUS)**


**APRIL :: 2026**

# DECLARATION

We undersigned hereby declare that the mini project report on "**CampUS**" submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by under supervision of **Prof. Ariyamol P G**, Assistant Professor, Dept of CSE. This submission represents our ideas in our own words and where ideas and words of others have been included. We have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a course for disciplinary action by the institute and/or the University and can also evoke panel action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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# CERTIFICATE

This is to certify that the report entitled "**CampUS**" submitted by **LIJO THOMAS (SJC23CS155), SOURAV S (SJC23CS203), SREENANDHU JACY (SJC23CS205), VAISAKH MANU (SJC23CS213)** to the APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering is a bonafide record of the mini project work carried out by them under my guidance and supervision.

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# Abbreviations

| | | |
|---|---|---|
| API | - | Application Programming Interface |
| JWT | - | JSON Web Token |
| ER | - | Entity Relationship |
| HTML | - | Hypertext Markup Language |
| JSON | - | JavaScript Object Notation |
| AI | - | Artificial Intelligence |
| UI | - | User Interface |
| UX | - | User Experience |
| DBMS | - | Database Management System |
| SQL | - | Structured Query Language |
| IDE | - | Integrated Development Environment |

# Abstract

In many educational institutions, communication among students, teachers, and administrators remains inefficient and fragmented across multiple platforms such as notice boards, emails, and messaging applications, resulting in delays and missed delivery of critical information including notices, examination schedules, timetables, and seating arrangements, which often leads to confusion and poor coordination. To address these challenges, the proposed system focuses on the development of CampUs, a centralized, secure, and intelligent campus communication and networking platform designed to streamline academic information exchange and campus interactions. The system provides role-based user registration and authentication for students, teachers, and administrators, enabling authorized access to real-time notices, announcements, and exam-related details with instant cloud-based updates. CampUs supports one-to-one and group communication between users and incorporates a peer connection mechanism to encourage academic collaboration and social interaction within the campus. To enhance usability and user support, the platform integrates an AI-powered chatbot that assists users by answering common queries, explaining notices, and guiding them through system features. Administrative functionalities such as user management, content moderation, and secure cloud storage ensure data integrity and system reliability. Overall, the proposed solution aims to overcome existing communication gaps by delivering a unified, interactive, and intelligent campus ecosystem that effectively combines real-time communication, exam information management, social connectivity, and AI-driven assistance.

# Chapter 1

# Introduction

In today's digital era, effective communication plays a vital role in academic institutions. However, most campuses still rely on fragmented systems such as notice boards, emails, and messaging applications, which often lead to delays and miscommunication. To overcome these challenges, a centralized digital solution is required to streamline communication and improve collaboration among students, faculty, and administrators. The proposed system, CampUs, is designed as an intelligent campus communication and information management platform that integrates real-time notifications, academic discussions, event management, and AI-based assistance into a single unified application. The system ensures efficient information delivery, enhances collaboration, and creates a connected campus environment.

## 1.1 Problem Statement

Campus communication systems are currently fragmented and inefficient. Important academic updates are often delayed or missed due to reliance on multiple platforms such as notice boards, emails, and messaging apps.
This lack of a centralized system leads to:

- Missed announcements
- Poor coordination
- Lack of collaboration among students and faculty

## 1.2 Objectives and Scope

The main objective of CampUs is to develop a centralized platform for campus communication and collaboration. The key objectives include:

- Providing real-time academic notifications
- Enabling secure role-based authentication
- Supporting communication through chats and forums
- Facilitating mentor–mentee interaction
- Enabling event management and participation
- Integrating AI for intelligent assistance

Scope:

The system focuses only on academic communication and collaboration. It does not handle marks, fees, or attendance systems.

## 1.3 Methodology Overview

The development of CampUs follows an iterative and modular approach:

- **Requirement Analysis** – Understanding user needs and system requirements
- **System Design** – Creating diagrams and architecture
- **Development** – Using React Native, Node.js, Supabase, and TypeScript
- **Integration** – AI chatbot and real-time features
- **Testing  Deployment** – Ensuring performance and reliability

# Chapter 2

# Literature Survey

## 2.1 Literature Referenced

### 2.1.1 Doe et al. (2020) – AI Chatbots in Education

The objective of this research is to explore the use of AI chatbots in educational environments to improve accessibility and user support. The methodology involves integrating chatbot systems into learning platforms and evaluating their effectiveness in answering student queries.

The study finds that AI chatbots provide instant responses, reduce faculty workload, and enhance user experience. They are capable of handling repetitive queries and guiding users through system functionalities. This improves efficiency and accessibility in educational systems [5].

A key strength of this research is its focus on automation and intelligent assistance. However, the effectiveness of chatbots depends on the quality of training data, and incorrect responses may affect user trust.

This research is highly relevant to CampUs as it supports the integration of an AI-powered assistant. CampUs uses similar concepts to provide intelligent guidance and improve user interaction within the platform.

### 2.1.2 Smith et al. (2019) – Modern Campus Communication Systems

The primary objective of this study is to analyze the evolution of campus communication systems and identify the limitations of traditional communication methods such as notice boards and emails. The methodology involves a comparative analysis of conventional and digital communication systems used in educational institutions. The study examines how centralized digital platforms can improve information dissemination and coordination among students and faculty.

The key findings indicate that traditional communication systems are inefficient due to delays and lack of accessibility. The research emphasizes the need for real-time notification systems and centralized platforms to ensure that important academic information reaches users instantly. The study also highlights the importance of role-based access control and secure data management for maintaining system integrity [1].

One of the major strengths of this work is its comprehensive analysis of communication gaps in academic environments. However, the study lacks implementation details and does not incorporate modern technologies such as AI-based assistance or mobile-first approaches. Additionally, it does not address user engagement and collaborative features.

This research is highly relevant to CampUs, as it provides a strong foundation for developing a centralized communication platform. CampUs extends this concept by integrating real-time messaging, AI-based assistance, and collaborative tools to create a more interactive and efficient academic ecosystem.

### 2.1.3 Kurniawan et al. (2019) – Design and Implementation of a Campus Social Networking System

The objective of this research is to design and implement a campus social networking system that facilitates academic collaboration among students. The methodology includes developing a web-based platform with features such as messaging, profile management, and content sharing. The system is tested to evaluate its effectiveness in improving communication.

The study finds that centralized platforms significantly improve interaction and collaboration among students. It enables users to share knowledge, discuss academic topics, and stay updated with campus activities. The system demonstrates improved communication efficiency compared to traditional methods [3].

The strength of this work lies in its practical implementation and focus on user interaction. However, it lacks real-time notification systems and does not integrate intelligent features such as AI-based assistance. It also does not include structured academic announcements or administrative control.

This research directly influences the development of CampUs. While adopting the concept of centralized networking, CampUs enhances it by adding real-time updates, AI chatbot support, and structured academic communication features.

### 2.1.4 Al-Rahmi et al. (2017) – Social Media for Collaborative Learning

This study investigates how social media platforms can be used to enhance collaborative learning among students. The methodology includes analyzing user behavior and engagement on social media platforms in educational contexts. The research evaluates how these platforms support knowledge sharing and interaction.

The findings indicate that social media significantly improves collaboration, communication, and learning outcomes. Students are more engaged when they actively participate in discussions and share information. The study highlights the role of interactive platforms in improving academic performance [4].

The strength of this research is its focus on user engagement and collaborative learning. However, social media platforms are not specifically designed for academic use, leading to issues such as distraction, lack of structure, and privacy concerns.

This study is relevant to CampUs as it supports the inclusion of discussion forums and communication features. CampUs improves upon this by providing a structured, secure, and academic-focused environment for collaboration.

### 2.1.5 Dabbagh et al. (2012) – Personal Learning Environments and Social Media

This research focuses on the integration of personal learning environments with social media to support self-regulated learning. The methodology includes analyzing how digital tools can enhance independent learning and collaboration.

The study finds that combining structured learning systems with social interaction tools improves learning outcomes. It allows students to manage their learning while interacting with peers [7].

The strength of this research is its emphasis on personalized learning. However, it lacks centralized control and does not provide a unified platform for academic communication.

This study is relevant to CampUs as it supports the idea of combining structured academic systems with collaborative tools, which CampUs implements effectively.

### 2.1.6 Hrastinski et al. (2009) – Online Learning as Participation

This study aims to understand online learning as a process of participation and interaction rather than passive content consumption. The methodology involves analyzing student engagement in online learning environments.

The findings suggest that active participation, communication, and interaction are essential for effective learning. The study emphasizes that communication tools play a critical role in enhancing student involvement [6].

The strength of this work lies in its theoretical foundation for online interaction. However, it does not provide a concrete system implementation or address real-time communication challenges.

This study is relevant to CampUs as it highlights the importance of participation and interaction. CampUs integrates chat systems, forums, and collaborative tools to promote active engagement among users.

### 2.1.7 Dawson et al. (2008) – Student Social Networks and Sense of Community

This study aims to explore the relationship between student social networks and their sense of community within educational institutions. The methodology involves analyzing student interactions and network structures using social network analysis techniques. The research focuses on how communication patterns influence student engagement and academic performance.

The findings reveal that students who are more connected within social networks tend to have a stronger sense of belonging and better academic outcomes. The study highlights the importance of communication platforms in fostering collaboration and peer learning [2].

A major strength of this research is its focus on the psychological and social aspects of communication. However, it does not provide a structured platform for academic communication and lacks technological implementation details. Additionally, it does not address issues such as data security and role-based access.

This study is relevant to CampUs as it emphasizes the importance of connectivity and collaboration among students. CampUs incorporates these concepts by providing discussion forums, chat systems, and mentor–mentee interactions to enhance student engagement and community building.

## 2.2 Existing Solutions

### 2.2.1 Moodle (2023) – Open-Source Learning Management System

The objective of Moodle is to provide a centralized platform for managing academic content, including courses, assignments, and assessments in educational institutions. It is designed to support structured learning by allowing teachers to upload materials, create quizzes, and track student performance. The methodology used in Moodle is based on a modular and web-based architecture, where different learning components such as courses, resources, and evaluations are integrated into a single system. It relies on a database-driven approach and supports plugin-based extensions for customization.

The key findings show that Moodle is highly effective in delivering structured academic content and managing learning activities efficiently. It allows institutions to maintain organized course materials and ensures accessibility for students. The major strengths of Moodle include its open-source nature, scalability, flexibility, and strong course management capabilities. However, it has certain limitations such as lack of real-time communication features, complex user interface for beginners, and absence of modern functionalities like AI-based assistance and instant notifications.

This system is relevant to CampUs as it highlights the importance of centralized academic systems. However, CampUs improves upon Moodle by integrating real-time communication, discussion forums, and intelligent assistance, thereby providing a more interactive and user-friendly campus communication platform.

### 2.2.2 Microsoft Teams (Microsoft, 2023) – Real-Time Communication and Collaboration Platform

The objective of Microsoft Teams is to provide a unified platform for real-time communication and collaboration through messaging, video conferencing, and file sharing. It is widely used in educational institutions for conducting online classes and facilitating group discussions. The system includes features such as chat, video meetings, file sharing, and collaboration tools. The methodology involves cloud-based communication services that enable real-time interaction and synchronization across multiple devices, ensuring seamless connectivity among users.

The key findings indicate that Microsoft Teams significantly enhances communication and collaboration among users. It provides efficient real-time interaction and supports teamwork through shared resources and communication channels. The major strengths of Microsoft Teams include its real-time communication capabilities, integration with other Microsoft services, and support for collaboration. However, it also has limitations such as a complex interface, high system resource usage, and lack of campus-specific structured features like centralized announcements and academic-focused notifications.

This application is relevant to CampUs as it demonstrates the effectiveness of real-time communication platforms. CampUs extends this concept by offering a simplified, campus-focused system that combines structured academic communication, real-time messaging, and AI-based assistance, making it more suitable for educational environments.

# Chapter 3

# System Study

## 3.1 Technologies

### 3.1.1 React Native

React Native is used as the primary framework for developing the CampUs mobile application. It enables the creation of cross-platform mobile apps using a single codebase, allowing the application to run efficiently on both Android and iOS devices. Its component-based structure supports reusable UI elements, which is useful in CampUs for building screens such as feeds, chats, notifications, profiles, and event pages with a smooth and consistent user experience.

### 3.1.2 Expo

Expo is used as the development platform for the CampUs mobile app. It simplifies the React Native development process by providing ready-made tools and APIs for tasks such as notifications, image handling, file access, and device features. In CampUs, Expo helps speed up development and makes it easier to manage and deploy the application across multiple platforms.

### 3.1.3 TypeScript

TypeScript is used to improve code quality and maintainability in the CampUs project. It adds static typing to JavaScript, which helps detect errors early during development and makes the code easier to understand and debug. In CampUs, TypeScript is especially useful because the application contains many modules such as authentication, messaging, event management, and team collaboration, all of which require structured and reliable code.

### 3.1.4 Supabase

Supabase is used as the main backend service for CampUs. It provides authentication, database access, real-time updates, and storage support, making it suitable for managing user accounts, chats, feeds, notifications, and uploaded media. In CampUs, Supabase helps maintain a scalable backend infrastructure while reducing the need to build everything from scratch.

### 3.1.5 Zustand

Zustand is used for state management in CampUs. It helps store and manage shared application data in a lightweight and efficient way, especially for states that need to be accessed across multiple screens or components. In CampUs, Zustand supports features like user session data, chat state, notifications, and UI-related state, helping the app remain responsive and organized.

### 3.1.6 Context API

The Context API is used in CampUs to manage global application state. It is useful for sharing information such as authentication status, theme preferences, and user-related data across the app without passing props through multiple components. This helps simplify state handling in parts of CampUs where global values are needed throughout the application.

### 3.1.7 Node.js

Node.js is used on the backend side of the CampUs project. It provides the runtime environment for the server-side logic that powers the AI extraction and communication features of the platform. In CampUs, Node.js supports backend processing tasks such as handling API requests, connecting services, and running Express-based routes for AI and admin-related operations.

### 3.1.8 Express

Express is used as the web framework for the CampUs backend server. It helps define routes, handle requests and responses, and organize server-side logic in a clean and structured way. In CampUs, Express is used for features such as chat processing, event extraction, poster extraction, and admin user management, making backend communication efficient and maintainable.

### 3.1.9 PostgreSQL

PostgreSQL is used as the relational database behind Supabase in CampUs. It stores structured application data such as user profiles, conversations, messages, events, projects, and notifications. In CampUs, PostgreSQL provides a reliable and scalable database foundation that supports real-time campus collaboration and secure data management.

## 3.2 Software Components

### 3.2.1 API Layer and Backend Services

The API layer and backend services ensure smooth communication between the frontend and backend. In the frontend, the API modules in the application centralize requests to Supabase and backend endpoints for features such as authentication, chat, feeds, events, projects, and notifications. On the backend, the Express server defines structured routes for tasks such as chat processing, event extraction, poster extraction, and admin user management. Middleware and server-level handlers manage CORS, request parsing, logging, environment validation, and error handling, ensuring secure, reliable, and maintainable data flow throughout the system.

### 3.2.2 Mobile App Frontend

The mobile app frontend is built using React Native and Expo, which provide the main user interface for students, faculty, and administrators. It is responsible for rendering screens, handling user interactions, and delivering a smooth cross-platform experience on Android, iOS, and web. This layer connects users to all core features of the platform through a responsive and modern interface.

### 3.2.3 Authentication and User Management

This component handles user sign-up, login, profile creation, and role-based access control. It ensures that users are properly identified as students, faculty, alumni, or admins, and that each role can access only the features permitted to them. User management also includes profile updates, avatar handling, and account settings.

### 3.2.4 Campus Feed

The campus feed acts as the central stream for announcements, notices, events, and academic updates. It allows users to view, like, and comment on posts, making it a shared communication space for the campus community. Admin and faculty moderation help keep the feed relevant and organized.

### 3.2.5 Chat and Messaging

The chat and messaging system supports direct communication between users as well as group conversations for projects and teams. It includes real-time message delivery, typing indicators, read receipts, reactions, and file attachments. This component is essential for quick collaboration and daily communication within the platform.

### 3.2.6 Discussions

The discussions module provides topic-based forums where users can exchange ideas, ask academic questions, and participate in structured conversations. It is designed to support categorized discussions so that messages remain easy to find and follow. This helps build a knowledge-sharing environment across departments and student groups.

### 3.2.7 Events and Reminders

This component manages campus events such as seminars, workshops, hackathons, and academic activities. Users can create, view, and register for events, while reminders and notifications help them stay informed about upcoming schedules. It also supports banners and discussion threads before and after events for better engagement.

### 3.2.8 Projects and Team Collaboration

The projects and team collaboration module allows students to create, join, and manage project teams, with features for team chat and coordination. It includes AI-based student recommendations that match project requirements with user profiles (skills, interests, and experience) to help project leaders select the most suitable members. Additionally, project-level mentorship is supported, enabling teams to connect with relevant mentors directly within the project for guidance and support.

### 3.2.9 Mentorship System

The mentorship system connects students with senior students, alumni, and faculty through AI-based mentor matching. It analyzes user profiles to recommend the most suitable mentors and supports both individual mentorship and project-based mentorship, where project teams can interact with mentors through dedicated chats for continuous guidance and collaboration.

### 3.2.10 Notifications

The notifications component keeps users informed about important actions such as messages, connection requests, event reminders, and team invitations. It combines push notifications and in-app alerts to make sure users do not miss important updates. This improves engagement and helps users respond quickly to relevant activity.

### 3.2.11 Admin Panel

The admin panel provides tools for managing users, posts, discussions, reports, and system-wide moderation. It helps administrators control access, review content, broadcast announcements, and monitor platform activity. This component is important for maintaining order, safety, and proper governance of the application.

### 3.2.12 AI Extraction Backend

The AI extraction backend supports automated processing for features such as poster reading, event extraction, and OCR-based data capture. It uses a Node.js and Express service with supporting libraries for image processing and text recognition. This backend extends the app's capabilities by reducing manual data entry and improving content extraction accuracy.

# Chapter 4

# System Design

## 4.1 Activity Diagram

The activity diagram shown in Figure 4.1 illustrates the overall workflow of the CampUs system, starting from user access to system functionalities. The process begins with the user opening the application and performing registration or login. Once authenticated, users are redirected to role-based dashboards such as student, faculty, alumni, or admin. Each role performs specific activities like viewing announcements, participating in discussions, joining events, sending messages, and using AI assistance. The system also handles background processes such as notifications, database updates, and activity logging. This diagram clearly represents the sequential flow of actions and decision points within the system.

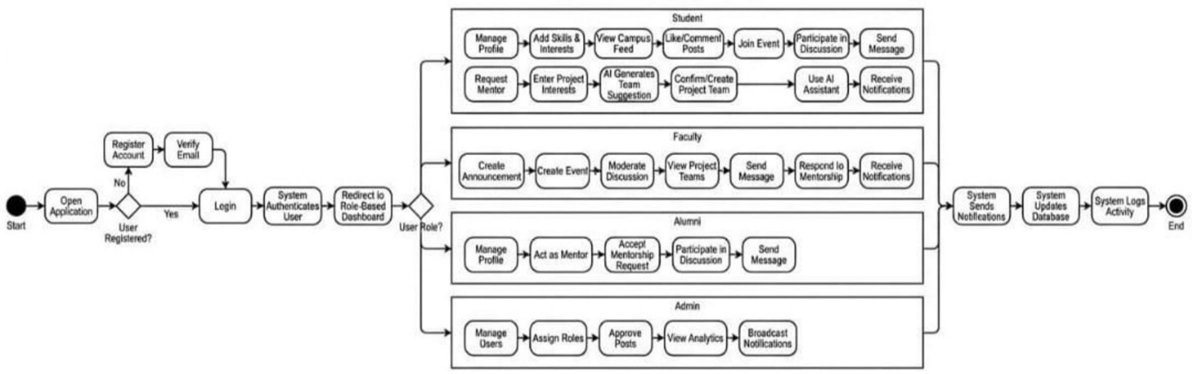



Figure 4.1: Activity diagram

## 4.2 Sequence Diagram

The sequence diagram shown in Figure 4.2 depicts the interaction between different components of the CampUs system, including the user, system, database, and AI service. The process starts with the user sending a login request, followed by credential validation through the database. Upon successful authentication, the system allows the user to access features such as viewing feeds and chats. The system retrieves data like announcements and messages from the database and displays it to the user. Additionally, the diagram shows how user queries are processed through the AI service, which generates responses and sends them back to the system. This diagram highlights the step-by-step communication flow and data exchange between system components.

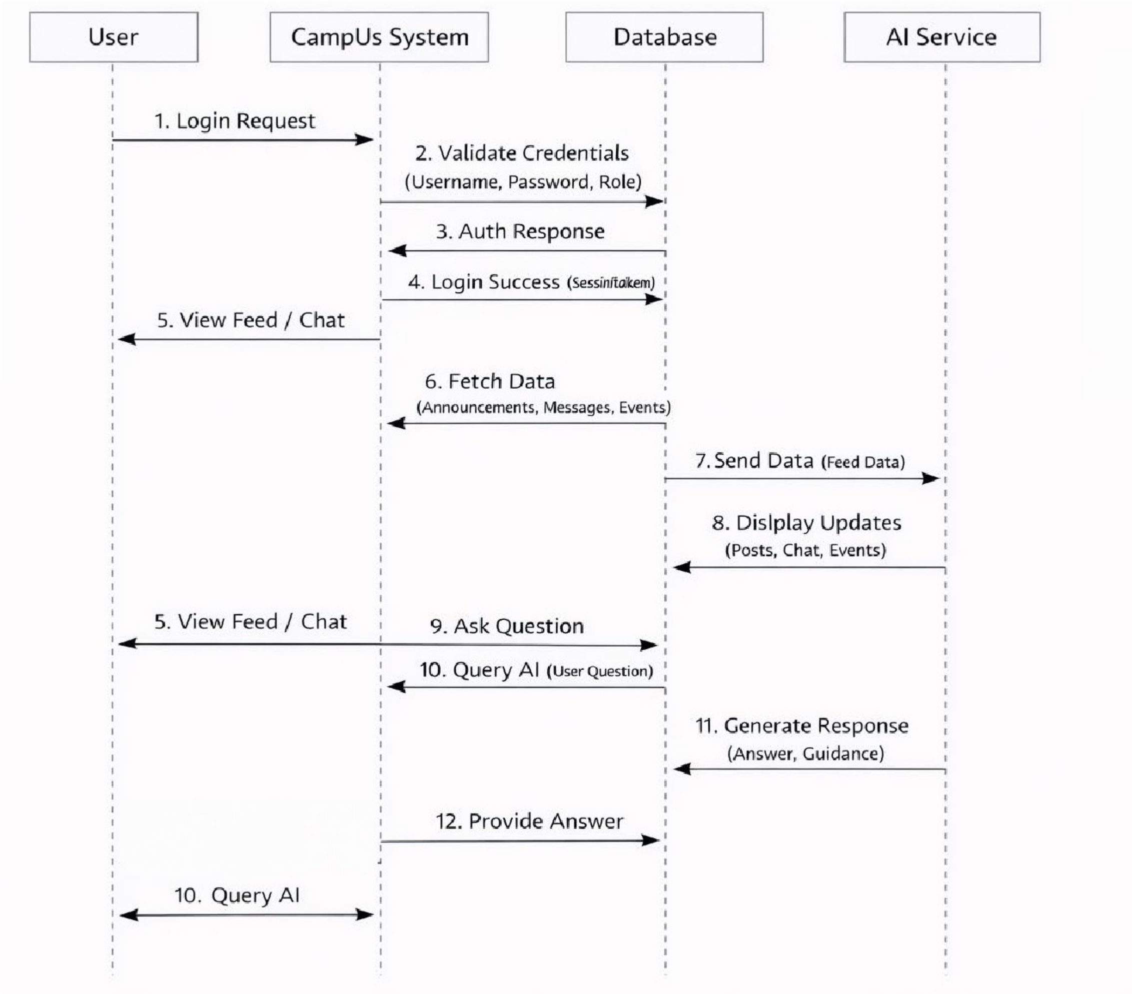



Figure 4.2: Sequence Diagram

## 4.3 ER Diagram

The ER diagram shown in Figure 4.3 represents the database structure of the CampUs system by illustrating various entities and their relationships. Key entities include User, Department, Post, Discussion, Event, Project, Skill, and MentorshipRequest. The User entity is central and connects to multiple entities through relationships such as creating posts, participating in discussions, joining projects, and requesting mentorship. Each entity contains attributes that define its properties, such as user details, post content, and event information. The relationships between these entities ensure proper data organization and integrity within the system. This diagram helps in understanding how data is stored, managed, and interconnected in the application.

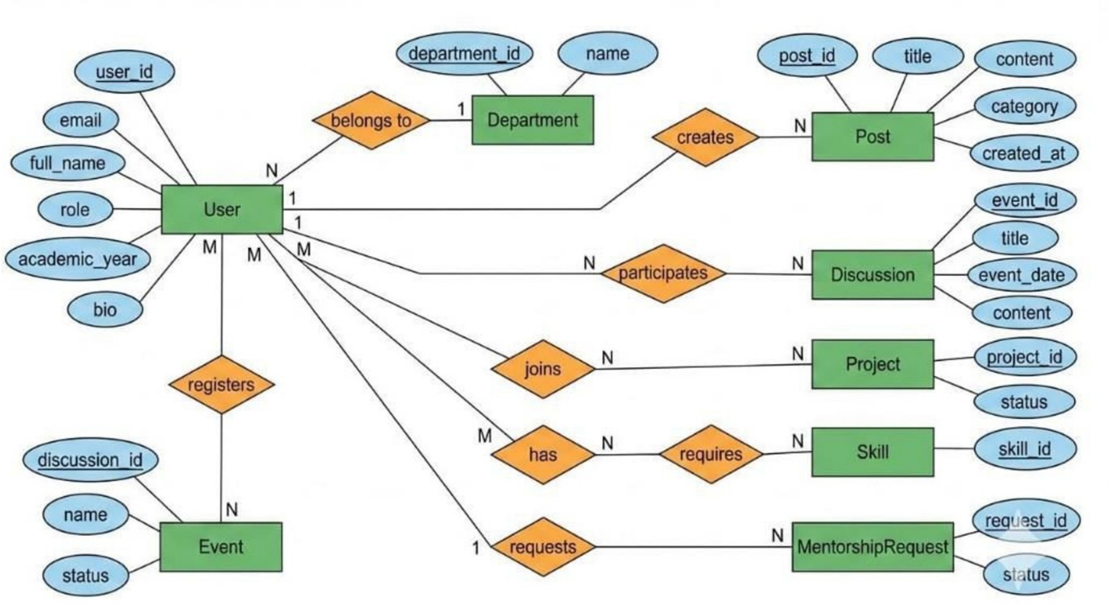



Figure 4.3: ER diagram

## 4.4 Use Case Diagram

The use case diagram shown in Figure 4.4 illustrates the interactions between different users and the CampUs system. The primary actors include students, faculty, alumni, and administrators, each having specific functionalities. Students can manage profiles, view announcements, participate in discussions, join events, and use AI assistance. Faculty members can create announcements, manage discussions, and interact with students. Alumni can act as mentors and participate in mentoring activities. Administrators have control over user management, role assignment, and content moderation. The diagram clearly defines the roles and responsibilities of each actor, providing a high-level view of system functionality.

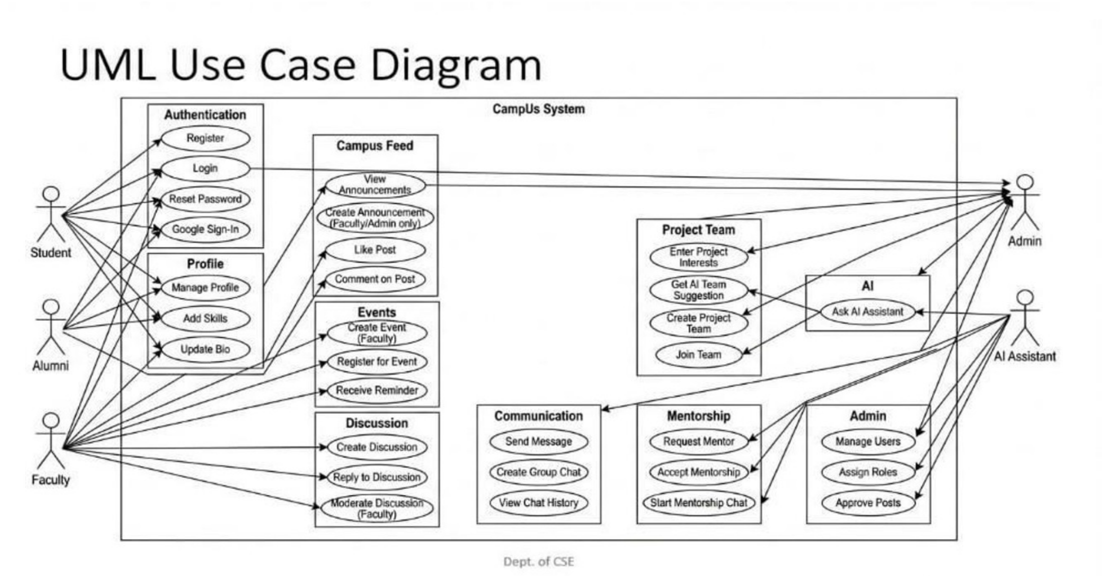



Figure 4.4: Use case diagram

# Chapter 5

# System Implementation

## 5.1 Modules

The CampUs system is implemented as a mobile application using React Native and Expo for the frontend, Supabase for authentication and database services, and Node.js with Express for backend processing. The system supports multiple user roles such as Student, Faculty, Alumni, and Admin. Each role is provided with specific functionalities through dedicated modules.

### 5.1.1 Authentication and User Management

The authentication module provides secure login and registration facilities for all users. Users can sign in using their email address and password. Supabase Auth is used to maintain user sessions and provide secure access to the system. After successful login, the system verifies the role of the user and redirects the user to the appropriate dashboard.

The module also supports password reset, password update, profile completion checking, and account suspension handling. If a user account is suspended, access to the application is blocked.

### 5.1.2 Profile Management

The profile management module allows users to manage their personal and academic information. Students can update details such as department, semester, year, skills, interests, and social media links. Faculty members can manage their department and designation details, while alumni can maintain information related to batch and academic background.

The module also supports profile picture upload, editing of personal information, and public profile viewing.

### 5.1.3 Student Module

The student module provides the main functionalities required by students. Students can browse the home feed, register for events, join project teams, communicate with faculty and alumni, and receive mentorship guidance.

The system also provides AI-based suggestions to help students find suitable mentors, collaborators, and academic opportunities according to their interests and academic background.

### 5.1.4 Faculty Module

The faculty module enables faculty members to create and manage academic posts, publish announcements, organize events, and mentor students. Faculty users can also review and approve inter-campus event submissions and respond to student queries.

This module improves communication between teachers and students and supports academic coordination within the institution.

### 5.1.5 Alumni Module

The alumni module allows alumni to remain connected with the institution. Alumni users can maintain their profile, participate in discussions, guide project teams, and provide mentorship for current students.

Through this module, students can connect with alumni for career guidance and professional networking.

### 5.1.6 Admin Module

The admin module provides complete control over the system. Administrators can manage users, change roles, suspend or reactivate accounts, create new accounts, moderate content, and monitor reports submitted by users.

The admin can also send announcements, view system statistics, and access logs for monitoring the activities taking place in the system.

### 5.1.7 Feed and Announcement Module

The feed module acts as the main dashboard of the application. It displays academic posts, announcements, event highlights, and other important information. Faculty and admin users can publish posts, while other users can view, like, comment on, and report posts.

The module helps in providing real-time communication and ensures that important information reaches all users effectively.

### 5.1.8 Event Management Module

The event module allows users to browse, create, and register for events. Events are categorized into upcoming, live, and completed events. Faculty and admin users can create events and upload event banners.

Students can register for events and receive reminders and notifications related to the event schedule.

### 5.1.9 Project and Team Formation Module

The project module supports the creation and management of project teams. Students can create project groups, invite members, accept or reject requests, and assign mentors to the project.

The team formation module also supports event team creation, allowing users to create and join teams for different academic and technical events.

### 5.1.10 Chat and Notification Module

The chat module provides real-time communication between users. It supports one-to-one chat, group chat, project chat, and mentorship chat. The notification module sends alerts for new messages, event reminders, project invitations, and announcements.

Realtime updates are implemented using Supabase services to ensure immediate delivery of notifications and messages.

### 5.1.11 AI Recommendation Module

The AI recommendation module provides intelligent suggestions to users. The system recommends mentors, collaborators, project teams, and events according to the profile, interests, and academic background of the user.

The module also supports AI-assisted event extraction and chatbot support through backend APIs.

### 5.1.12 InterCampus Module

The InterCampus module supports cross-college event participation. Users can submit event details using website links or posters. The system extracts event information using OCR, QR code detection, and AI-based processing.

Faculty and admin users can verify these submissions before they are published in the application.

# Chapter 6

# Experimental Results

## 6.1 Results and Discussions

The CampUS system was executed on Android devices and Windows systems using React Native, TypeScript, Node.js, Supabase, and PostgreSQL. The development and testing of the application were carried out using Visual Studio Code and Antigravity. The system was tested for all major modules including authentication, discussion forum, project management, profile management, event handling, messaging, and administration.

The input data, module implementation, and execution results are illustrated in the following screenshots. Each screenshot demonstrates the successful working of a particular module in the CampUS application.

## 6.2 System Execution Output

### 6.2.1 Login Module

The login module provides secure access to the CampUS application. Users can log in using their institutional email and password. After successful authentication, the system verifies the role of the user and redirects them to the appropriate dashboard. If the account is suspended or the profile is incomplete, the user is redirected accordingly.

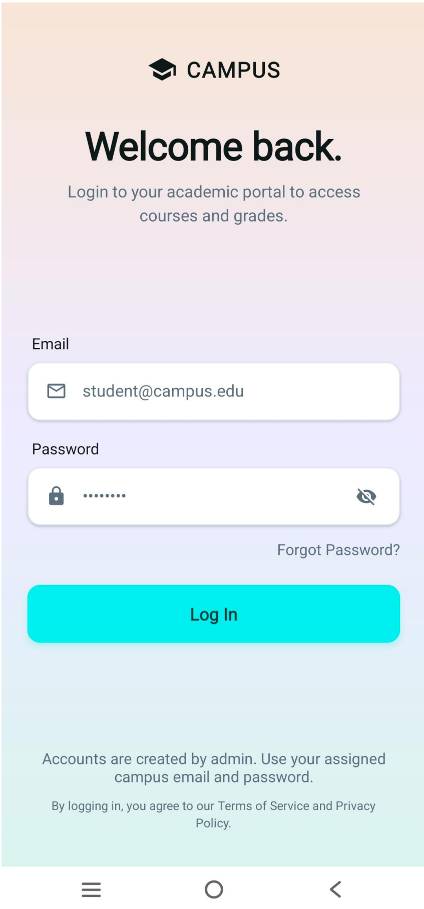



Figure 6.1: Login Screen

### 6.2.2 Discussion Module

The discussion module allows users to interact with each other by creating posts, asking questions, and participating in academic discussions. Users can view discussion topics, reply to posts, and contribute to knowledge sharing within the platform.

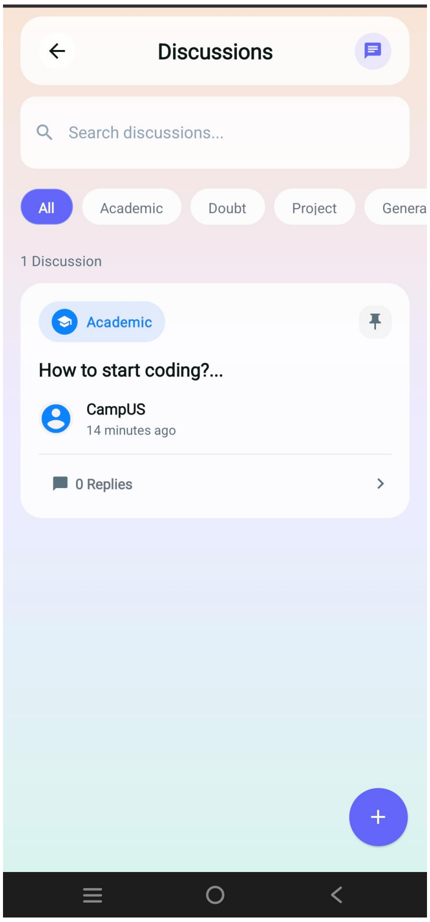



Figure 6.2: Discussion Forum

### 6.2.3 Project Module

The project module supports project team creation and management. Users can browse available projects, join project teams, and view complete project details including required skills, team members, and mentor information.

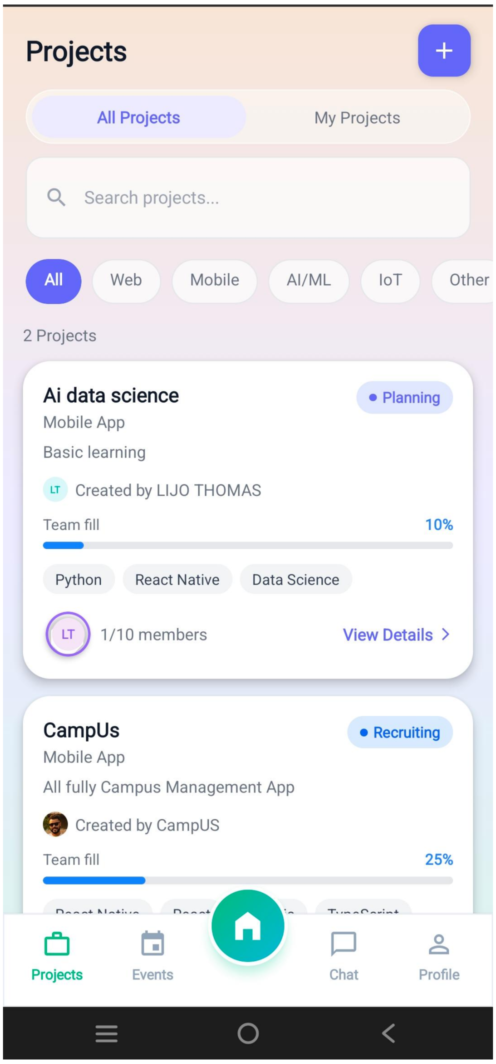



Figure 6.3: Project Team Formation

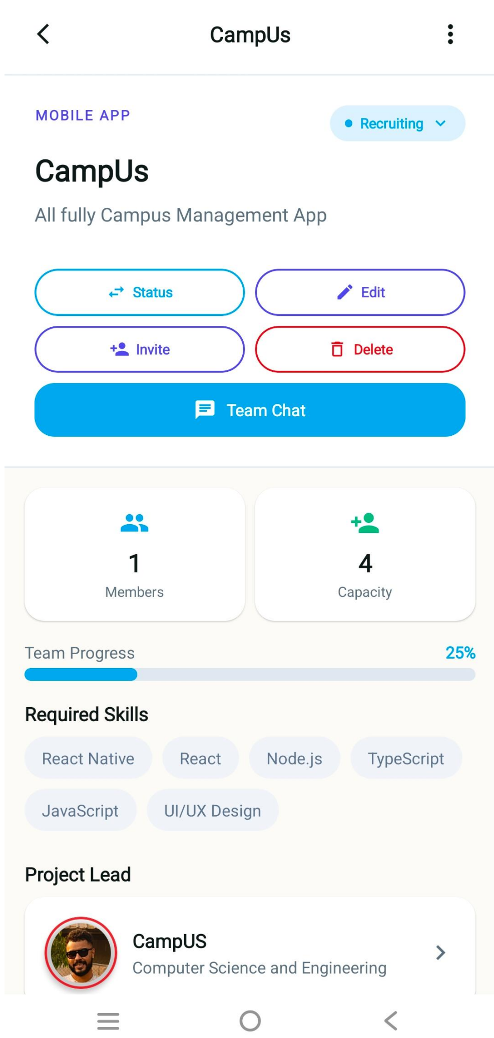



Figure 6.4: Project Details

### 6.2.4 Profile Module

The profile module allows users to manage their personal, academic, and professional details. Users can update their profile picture, department, skills, interests, and social links. The system also provides public profile viewing for other users.

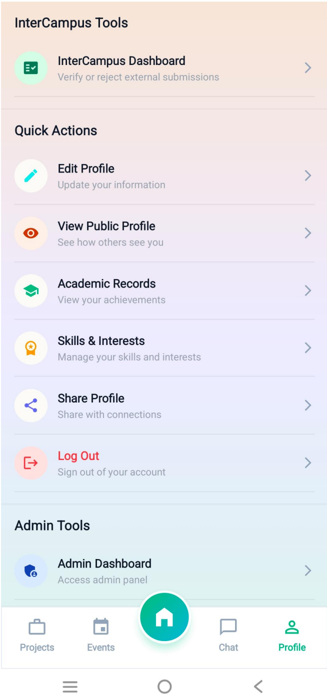



Figure 6.5: Profile Dashboard

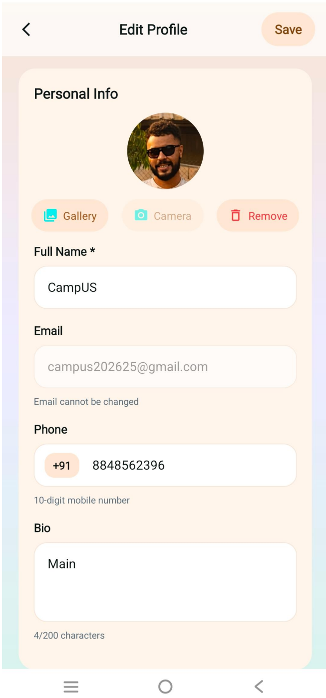



Figure 6.6: Edit Profile

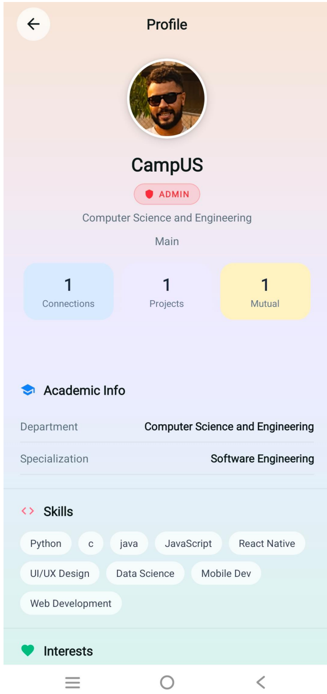



Figure 6.7: Public Profile

### 6.2.5 Dashboard Module

The dashboard module acts as the home page of the application. It provides quick access to announcements, projects, discussions, events, and other important features. The dashboard also displays recent activity and upcoming events.

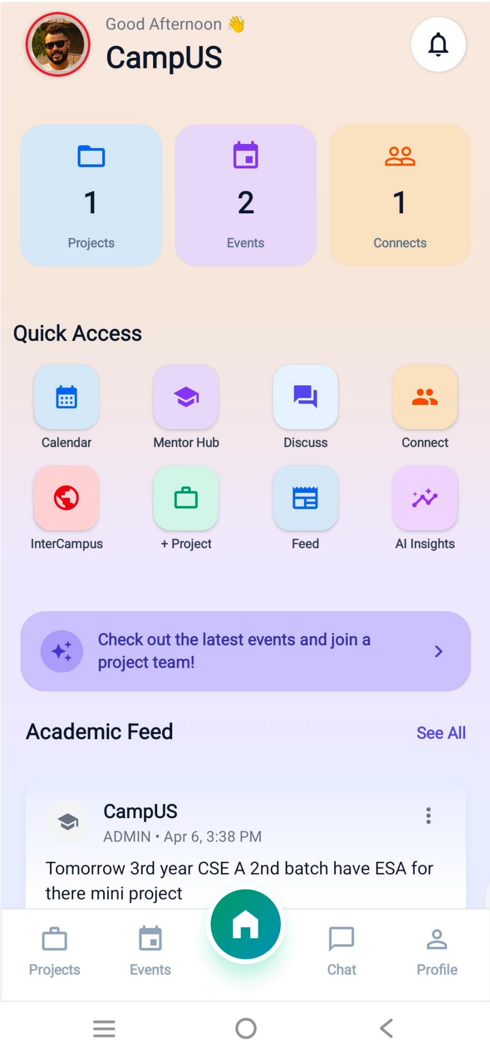



Figure 6.8: Home Dashboard

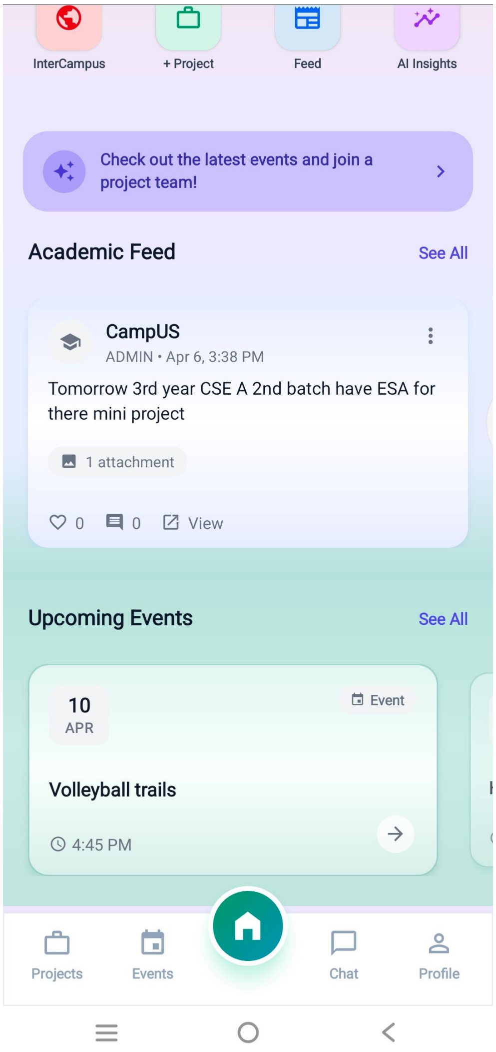



Figure 6.9: Main Dashboard

### 6.2.6 Feed Module

The campus feed module provides a centralized location for announcements, notices, academic updates, and community posts. Users can view posts from different categories such as announcements, exams, events, and general discussions. The feed allows users to like, comment, and share posts, fostering active engagement and information sharing across the campus community.

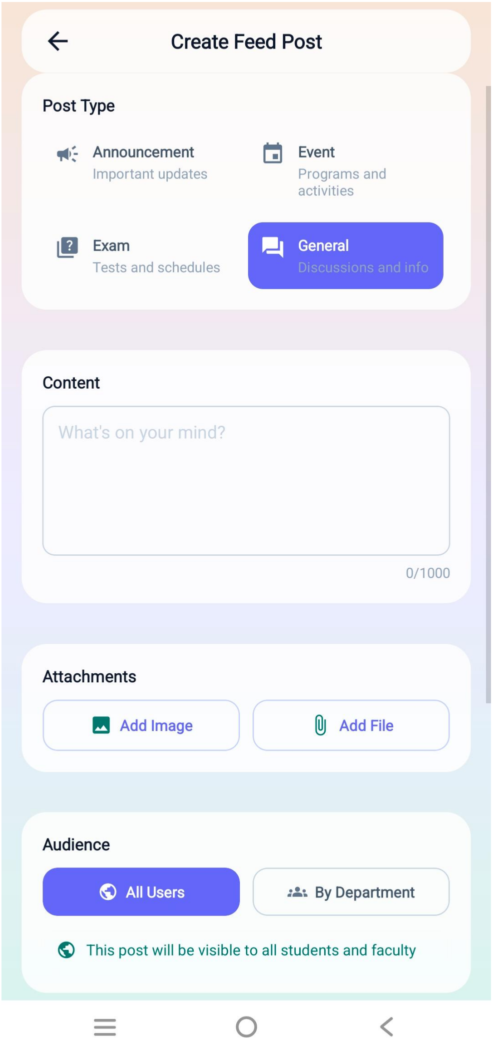



Figure 6.10: Create Feed Post

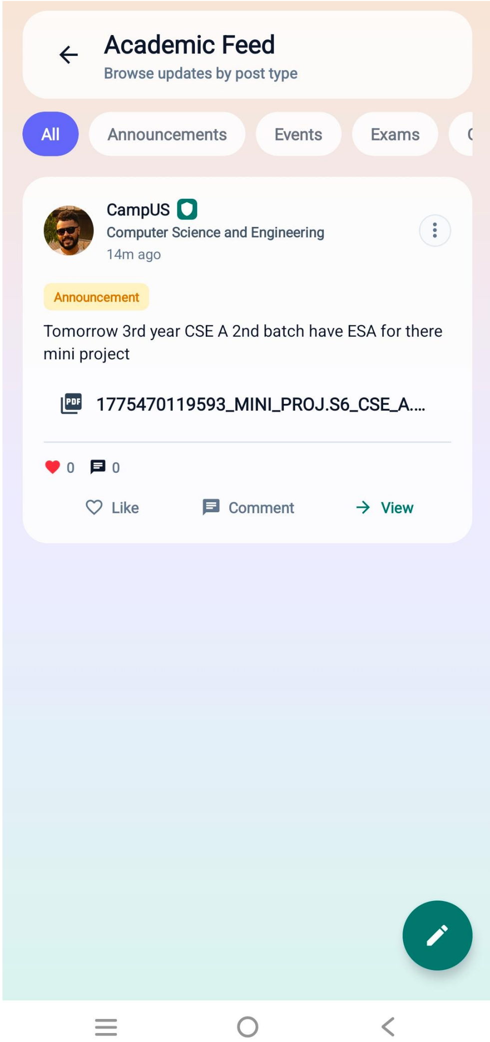



Figure 6.11: Academic Feed

### 6.2.7 Event Module

The event module allows users to browse, register, and view details of academic and technical events. Events are categorized based on date and status. Users can register for events and view additional details such as venue, time, and description.

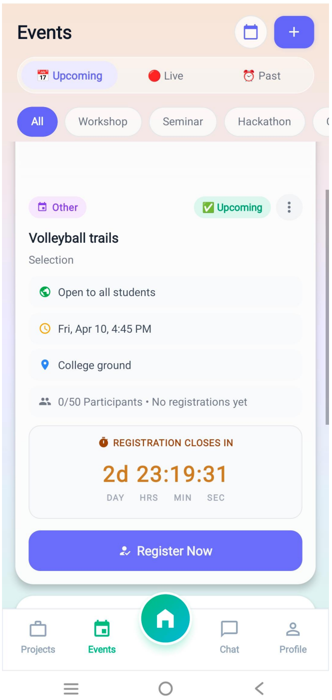



Figure 6.12: Event Management

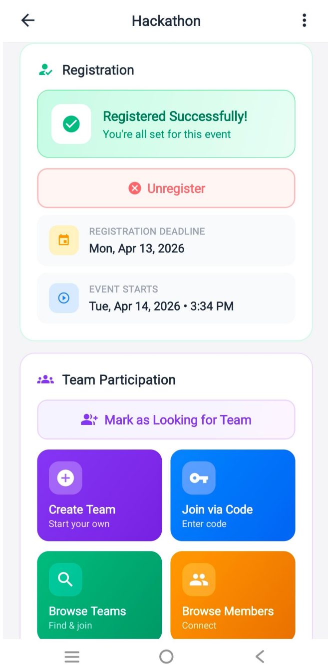



Figure 6.13: Event Registration

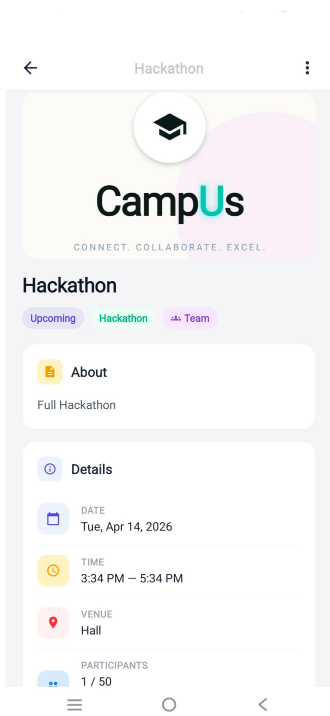



Figure 6.14: Event Details

### 6.2.8 Messaging Module

The chat and messaging interface shown below allows users to send direct messages, participate in group conversations, and collaborate through real-time communication. The interface displays typing indicators when other users are composing messages, shows read receipts to confirm message delivery, and supports emoji reactions and file attachments. Users can also delete, forward, and reply to messages, making the chat system highly functional for both personal and team communication.

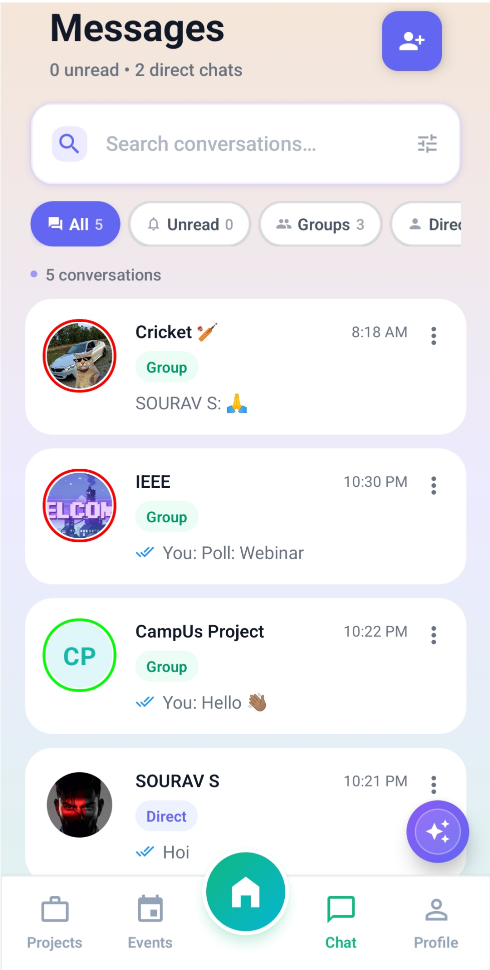



Figure 6.15: Chat Module

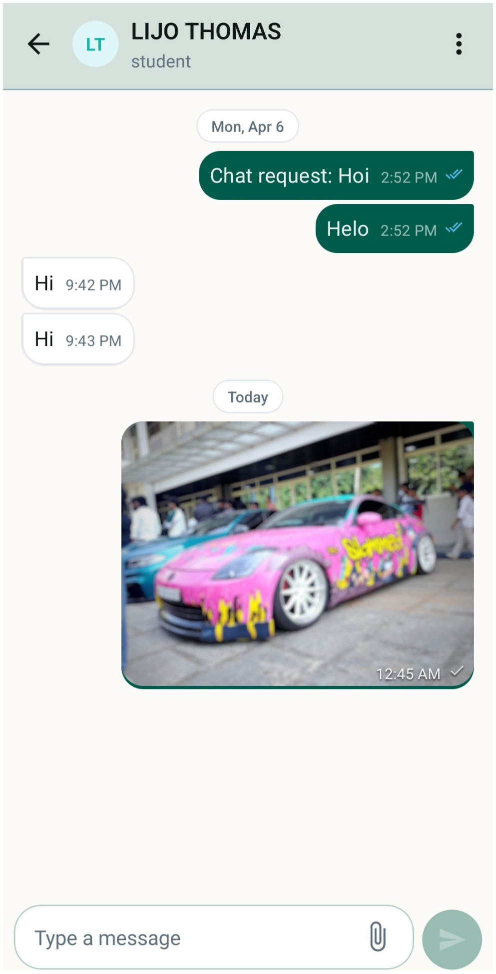



Figure 6.16: Direct Message Module

### 6.2.9 User Discovery Module

The user discovery module helps users search and connect with students, faculty, and alumni. Users can browse available profiles and send connection requests.

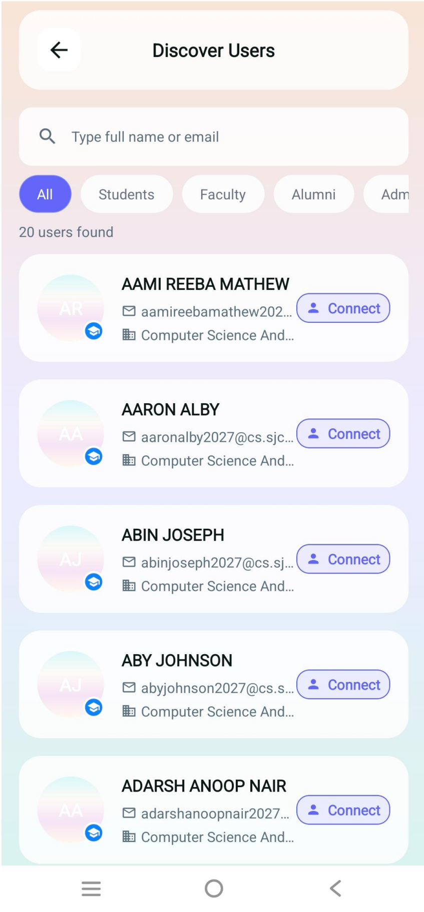



Figure 6.17: User Discovery

### 6.2.10 Calendar Module

The calendar module displays upcoming events and reminders in a structured format. Users can view the date, time, and event details directly from the calendar interface.

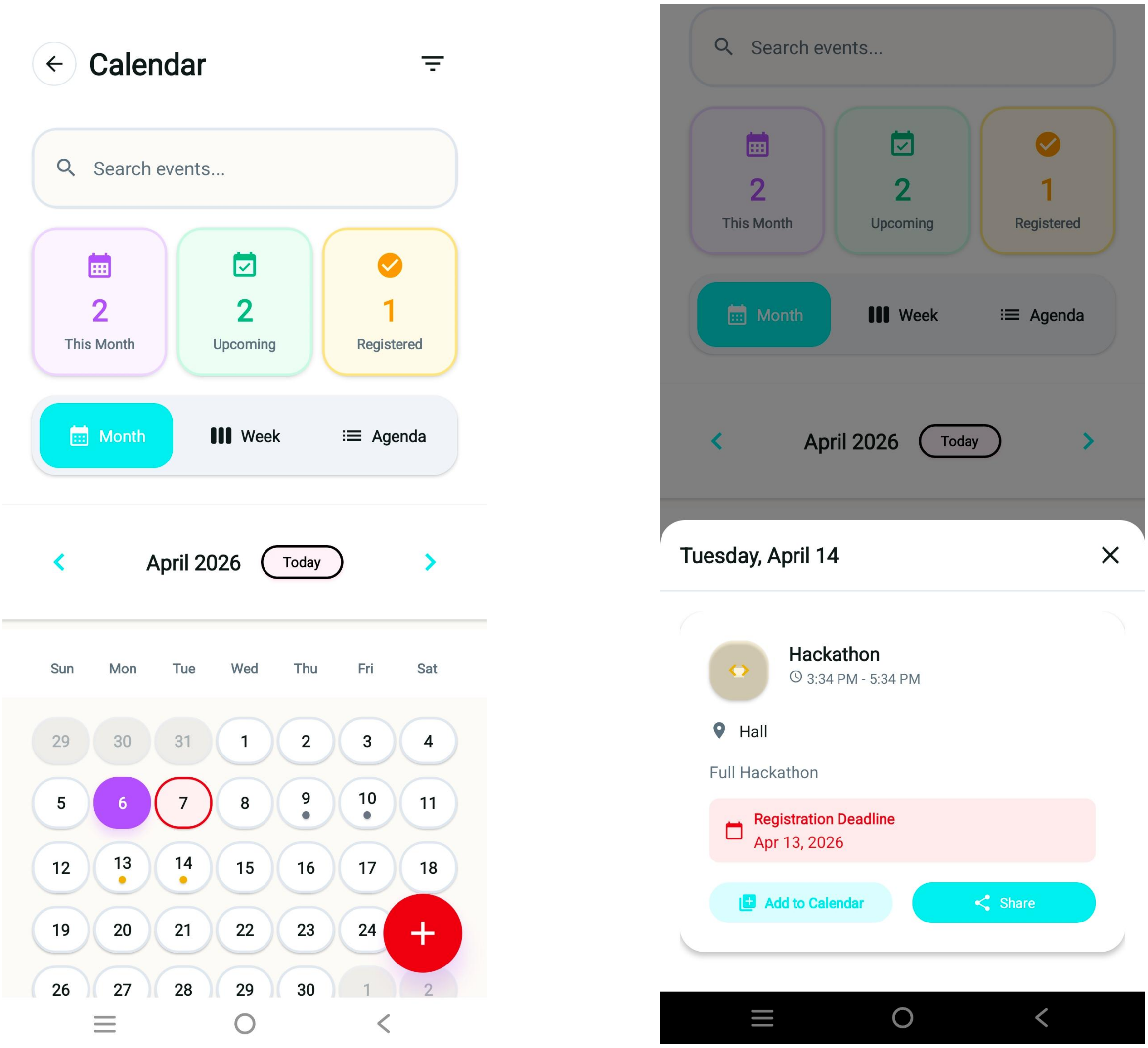



Figure 6.18: Calendar and Event Tracking

### 6.2.11 Admin Module

The admin dashboard shown, provides administrators with a centralized control panel for managing the platform. It displays key metrics such as total users, active discussions, pending reports, and system health. Administrators can manage users, monitor reports, review activities, and maintain the overall functionality of the application. The dashboard includes quick access buttons to user management, content moderation, report handling, and system settings, enabling efficient platform administration.

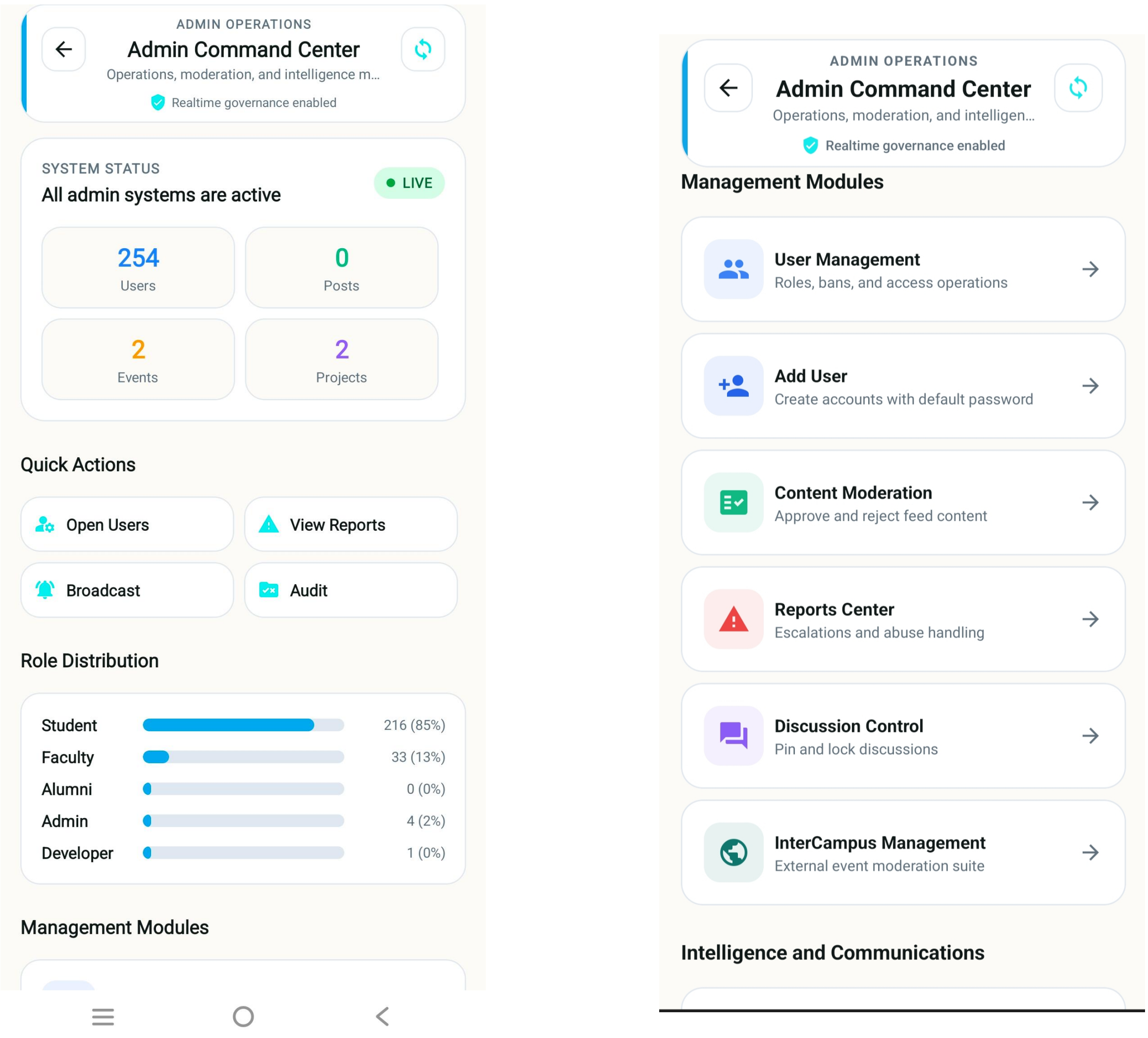



Figure 6.19: Admin Dashboard

# Chapter 7

# Conclusion

The CampUs project was developed with the primary objective of creating a centralized academic communication and collaboration platform to overcome the limitations of fragmented systems such as notice boards, WhatsApp groups, and other unofficial channels. The system plays a significant role in ensuring efficient and real-time information flow within the campus by integrating secure role-based authentication, centralized announcements, communication tools, and mentoring features. By bringing all academic interactions into a single unified platform, CampUs enhances coordination, reduces information delays, and improves overall campus productivity.

The project successfully achieved its goals by implementing key features such as real-time communication and discussion forums, mentor–mentee guidance systems, and secure user management using technologies like Node.js, React Native, Supabase, and TypeScript, with development carried out using Visual Studio Code and Antigravity as code editors. Despite its effectiveness, challenges such as ensuring scalability, real-time synchronization, and AI integration were encountered.

Future enhancements can include advanced AI-based recommendations, multi-campus integration, and improved analytics for better decision-making. Overall, CampUs stands as a modern, intelligent, and secure solution for digital academic communication and collaborative learning.

# Reference


[1] A. Doe, "AI Chatbots in Education," *International Journal of Computer Applications*, vol. 182, no. 45, pp. 10–15, 2020.

[2] Smith, *Modern Campus Communication Systems*, 1st ed., New York, NY, USA: Springer, 2019.

[3] A. B. S. Kurniawan and R. Hidayat, "Design and Implementation of a Campus Social Networking System for Academic Collaboration," in *IEEE International Conference on Information Technology Systems and Innovation*, pp. 151–156, 2019.

[4] M. Al Rahmi and A. M. Zeki, "A model of using social media for collaborative learning to enhance learners' performance," *IEEE Access*, vol. 5, pp. 5266–5278, 2017.

[5] N. Dabbagh and A. Kitsantas, "Personal learning environments, social media, and self-regulated learning," *Internet and Higher Education*, vol. 15, no. 1, pp. 3–8, 2012.

[6] S. Hrastinski, "A theory of online learning as online participation," *Computers & Education*, vol. 52, no. 1, pp. 78–82, 2009.

[7] S. Dawson, "A study of the relationship between student social networks and sense of community," *Educational Technology & Society*, vol. 11, no. 3, pp. 224–238, 2008.

[8] "React Native Documentation", [Online], Available: https://reactnative.dev

[9] "Node.js Documentation", [Online], Available: https://nodejs.org/en/docs

# Annexure A

ST.JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, - PALAI - AUTONOMOUS

CSD 334: MINI PROJECT

# CampUs

Final Review

Date:07/04/2026

Guided By,
ARIYAMOL P.G
Assistant Professor
Department Of Computer Science & Engineering

Presented By,
49 : LIJO THOMAS (SJC23CS155)
66 : SOURAV S (SJC23CS203)
67 : SREENANDHU JACY (SJC23CS205)
70 : VAISAKH MANU (SJC23CS213)





## Course Outcome

Course Outcomes: After the completion of the course the student will be able to;

| CO# | CO |
|---|---|
| CO1 | Identify technically and economically feasible problems (Cognitive Knowledge Level: Apply) |
| CO2 | Identify and survey the relevant literature for getting exposed to related solutions and get familiarized with software development processes (Cognitive Knowledge Level: Apply) |
| CO3 | Perform requirement analysis, identify design methodologies and develop adaptable & reusable solutions of minimal complexity by using modern tools & advanced programming techniques (Cognitive Knowledge Level: Apply) |
| CO4 | Prepare technical report and deliver presentation (Cognitive Knowledge Level: Apply) |
| CO5 | Apply engineering and management principles to achieve the goal of the project (Cognitive Knowledge Level: Apply) |





## CONTENTS







## Introduction

- Campus communication spread across platforms
- Students lack unified digital system
- Information delays cause confusion
- CampUs App centralizes campus networking
- It helps in the development of online social networks by connecting a user's profile with those of other individuals with similar interests





## Problem Statement

- Campus communication is inefficient and fragmented.
- Academic notices lack real-time delivery.
- Students miss critical academic updates.
- No centralized platform for campus interaction.

### Why Solution is needed!

- Ensures timely delivery of academic information.
- Reduces confusion and missed campus updates.
- Improves communication among students and faculty.
- Enhances efficiency through centralized digital access.





## Objective of the Project

- Centralize campus communication and networking
- Deliver real-time academic notifications
- Integrate AI for intelligent user support
- Connect users with shared interests
- Encourage collaboration through campus activities
- Enable meaningful peer engagement digitally
- Build inclusive student interest communities

# Existing System

**Basic campus communication system**

**Features:**
- Creates accounts with personal details and interests.
- Teachers can post notices and images.
- Separate pages for notices and images.
- Students can view updates from teachers.

**Limitations:**
- •Communication flow is top-down.
- •Platform lacks student-to-student networking features.
- •The scope is limited to posting and viewing announcements.
- •Basic social networking concept is not fully realized.





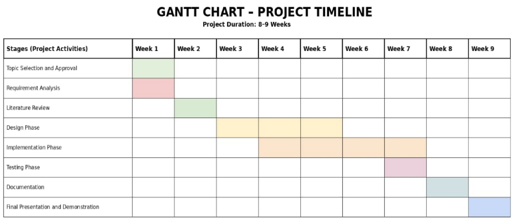
## Proposed System

- •**CampUs App:** Centralized campus communication system
- •**Interest Matching:** Connects users with shared interests
- •**Secure Access:** Role-based authentication and authorization
- •**Real-Time Updates:** Cloud-based notification delivery
- •**AI Support:** Chatbot for academic assistance
- •**Technology Stack:** React Native, Firebase, AI
- •**Features:**
  - •**One-to-One Messaging:** Secure student and faculty communication
  - •**Group Discussions:** Class and department-based chat groups
  - •**Event Management:** Campus events with schedules and reminders
  - •**Search & Filters:** Quick access to notices and users





## Gantt Chart






# Literature Survey

| Sl. No | Paper / System / Year | Focus Area | Methodology / Technology | Key Features | Limitations Identified | Relevanceto CampUs |
|---|---|---|---|---|---|---|
| 1 | Google Classroom (2014-2025) | Academic Course Management | Cloud-based LMS, Google Cloud | Assignment posting, announcements, classroom communication | No AI-based student matching; limited peer collaboration features | Highlightsneed for AI-driven collaboration & mentoring system |
| 2 | Microsoft Teams for Education | Institutional Communication | Cloud collaboration platform, real-time messaging | Group chat, video meetings, file sharing | Complex UI; not academic-interest focused; no AI team formation | Supports need for simplified academic-only communication platform |
| 3 | Moodle LMS | Learning Management System | Open-source LMS, modular architecture | Course management, forums, grading | No real-time chat; limited AI integration | Shows importanceof modular academic system withscalable backend |



| Sl. No | Paper / System / Year | Focus Area | Methodology / Technology Used | Key Features | Limitations Identified | Relevance to CampUs |
|---|---|---|---|---|---|---|
| 4 | LinkedIn (Professional Networking) | Skill-based Matching | AI-based recommendation engine | Skill endorsements, professional networking | Not academic-focused; no project-based team formation | Validates AI-based skill matching concept |
| 5 | Discord (Community Platform) | Real-time Communication | WebSocket-based messaging | Topic-based channels, role permissions | No academic moderation system; informal environment | Inspires structured discussion + role-based moderation |



# Module Description

1. User Authentication & Role Management Module
- Validates user login credentials using Supabase Authentication
- Supports Email/Password
- Identifies user roles (Student / Alumni / Faculty / Admin)
- Restricts unauthorized access using role-based access control
- Redirects users to role-specific dashboards

2. Academic Profile Management Module
- Allows users to create and update academic profiles
- Stores department, academic year, bio, and profile picture
- Enables adding skills and interests as tags
- Maintains project preferences for collaboration*
- Supplies profile data for AI-based recommendations and matching*

3. <u>Campus Feed & Announcement Module</u>
- Provides a centralized academic announcement feed
- Allows only Faculty/Admin to post official updates
- Displays posts in chronological order
- Supports likes, comments, and moderation
- Categorizes posts to prevent non-academic misuse

4. <u>Events & Academic Activities Module</u>
- Enables Faculty/Coordinators to create academic events
- Displays events in calendar and feed view
- Allows students to register/unregister for events
- Sends automated reminders using push notifications
- Provides pre-event discussion and post-event discussion space



5. <u>Communication & Discussion Module</u>
- Provides topic-based academic discussion forums
- Supports one-to-one and group chat messaging
- Enables real-time message delivery using Supabase sync
- Includes typing indicators and read receipts
- Supports cross-department collaboration

6. <u>Mentor–Mentee Guidance Module</u>
- Allows seniors and alumni to register as mentors
- Enables juniors to request mentorship
- Creates private mentoring chats upon acceptance



7. <u>Notification & Alert Module</u>
- Sends real-time push notifications
- Delivers event reminders and chat alerts
- Supports in-app notifications
- Allows notification preference settings

8. <u>Admin & Faculty Control Module</u>
- Allows user and role management
- Supports post approval and moderation
- Enables banning/suspending users
- Provides system analytics dashboard
- Broadcasts institution-wide announcements



## Database Design

| • Users | field |
|---|---|
| user_id | Varchar(50) (PK) |
| name | Varchar(100) |
| email | Varchar(100)(Unique) |
| password | Varchar(255) |
| role | Varchar(20) |
| department | Varchar(100) |
| academic_year | Int |
| bio | Text |
| profile_picture | Varchar(255) |
| is_verified | Boolean |
| status | Varchar(20) |
| created_at | Timestamp |

| • Skills | type |
|---|---|
| skill_id | Int (PK) |
| skill_name | Varchar(100) |

| • User_Skills | type |
|---|---|
| user_id | Varchar(50 (PK, FK) |
| skill_id | Int (PK, FK) |

| • Interests | type |
|---|---|
| interest_id | Int (PK) |
| interest_name | Varchar(100) |

| • User_Interests | type |
|---|---|
| user_id | Int (PK) |
| user_id | Varchar(50) (FK) |
| created_at | Timestamp |

| • Posts | type |
|---|---|
| post_id | Int (PK) |
| user_id | Varchar(50) (FK) |
| created_at | Timestamp |

| • Post_Likes | type |
|---|---|
| post_id | Int (PK, PK, FK) |
| user_id | Varchar(50) (FK) |
| created_at | Timestamp |

| • Comments | type |
|---|---|
| comment_id | Int (PK) |
| post_id | Varchar(50) (FK) |
| created_..1 | Timestamp |

| • Event_Registrations | type |
|---|---|
| event_id | Int (PK, FK) |
| user_id | Varchar(50) (FK) |

| • Forums | type |
|---|---|
| forum_id | Int (PK) |
| title | Varchar(200) |
| description | Text |
| event_date | Date |
| event_time | Time |
| location | Varchar(150) |
| created_at | Timestamp |

| • Events | type |
|---|---|
| event_id | Int (PK) |
| created_by | Varchar(50) (FK) |
| title | Varchar(200) |
| description | Text |
| event_date | Date |
| event_time | Time |
| location | Varchar(150) |
| created_at | Timestamp |

| • Chats | type |
|---|---|
| chat_id | Int (PK) |
| chat_type | Varchar(20) |

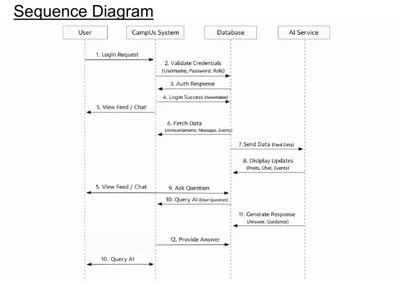
| • Chat_Members | type |
|---|---|
| chat_id | Int (PK, FK) |
| user_id | Varchar(50) (FK) |
| role_in_team | Varchar(50) |
| sent_at | Timestamp |

| • Mentor_Requests | type |
|---|---|
| request_id | Int (PK) |
| mentee_id | Varchar(50) (FK) |
| mentor_id | Varchar(50) (FK) |
| status | Varchar(20) |
| requested_at | Timestamp |

| • Project_Teams | type |
|---|---|
| team_id | Int (PK) |
| project_title | Varchar(200) |
| created_by | Varchar(50) (FK) |
| created_at | Timestamp |

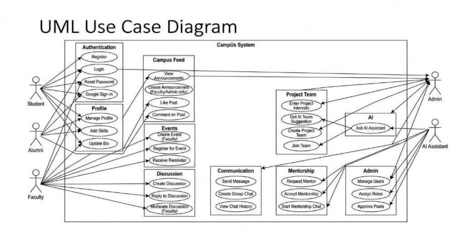


<u>Sequence Diagram</u>

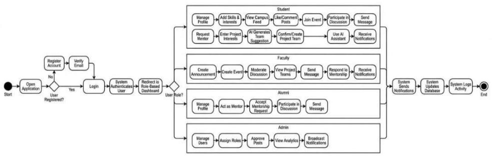





## <u>Use Case Diagram</u>

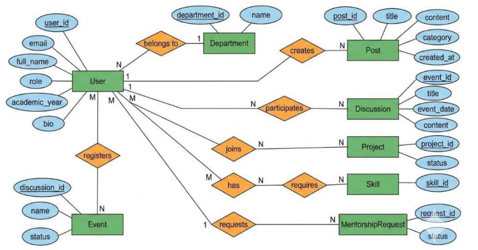


## Activity Diagram





## ER Diagram






## Conclusion

CampUs successfully proposes a centralized academic communication and collaboration platform designed to eliminate fragmented systems such as notice boards,WhatsApp groups and unofficial channels.

- The system integrates:
- Secure role-based authentication
- Centralized academic announcements
- Real-time communication and discussion forums
- Mentor–mentee guidance system



- Secure and scalable architecture
- Real-time synchronization
- Structured moderation and academic integrity
- Efficient collaboration within the campus ecosystem

CampUs provides a scalable foundation that can be extended in the future with:

- Institutional integrations
- Cross-campus collaboration features
- In conclusion, CampUs offers a modern, intelligent, and secure solution for digital academic communication and collaborative learning



## References

[1] IEEE Computer Society, IEEE Standard for Software Documentation, IEEE Std1063-2001, 2001.

[2] Smith, Modern Campus Communication Systems, 1st ed., New York, NY, USA:Springer, 2019.

[3] S. Dawson, “A study of the relationship between student social networks and senseof community,” Educational Technology & Society, vol. 11, no. 3, pp. 224–238, 2008.

[4] A. B. S. Kurniawan and R. Hidayat, “Design and Implementation of a Campus So-cial Networking System for Academic Collaboration,” in IEEE International Con-ference on Information Technology Systems and Innovation, pp. 151–156, 2019.

[5] M. Al Rahmi and A. M. Zeki, “A model of using social media for collaborativelearning to enhance learners’ performance,” IEEE Access, vol. 5, pp. 5266–5278,2017.

[6] A. Doe, “AI Chatbots in Education,” International Journal of Computer Applica-tions, vol. 182, no. 45, pp. 10–15, 2020.



Dept. of Computer Science & Engineering

THANK YOU

# VISION & MISSION OF THE DEPARTMENT

## Vision

To evolve as a school of computing with globally reputed Centre's of excellence and serve the changing needs of the industry and society.

## Mission

- The department is committed in bringing out career-oriented graduates who are industry ready through innovative practices of teaching and learning process
- To nurture professional approach, leadership qualities and moral values to the graduates by organizing various programs periodically
- To acquire self-sustainability and serve the society through research and consultancy